

LA 4.2

$$\begin{array}{l}
 p(x) = a_4 x^4 + a_3 x^3 + a_2 x^2 + a_1 x + a_0 = p(x) \\
 p(0) \quad \left(\begin{array}{ccccc|c} 0 & 0 & 0 & 0 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 & 2 \\ 16 & 8 & 4 & 2 & 1 & -3 \end{array} \right) \\
 p(1) \\
 p(3)
 \end{array}$$

A

Pro hledání $v = (a_4, a_3, a_2, a_1, a_0)^T$ platí: $Av = \begin{pmatrix} 1 \\ 2 \\ -3 \end{pmatrix}$

a $v \perp \text{Ker } A$

$$\begin{array}{c}
 \updownarrow \\
 v \in \text{Im } A^T \\
 \updownarrow
 \end{array}$$

$$\exists w \in \mathbb{R}^3: v = A^T w$$

$$A(A^T w) = \begin{pmatrix} 1 \\ 2 \\ -3 \end{pmatrix}$$

$$\left(\begin{array}{ccc|c} 1 & 1 & 1 & 1 \\ 1 & 5 & 31 & 2 \\ 1 & 31 & 341 & -3 \end{array} \right)$$

$$w = \begin{pmatrix} 1/10 \\ 1 \\ -1/10 \end{pmatrix}$$

$$A^T \cdot w = \frac{1}{5} \begin{pmatrix} -3 \\ 1 \\ 3 \\ 4 \\ 5 \end{pmatrix}$$

$$p(x) = \frac{1}{5} (-3x^4 + x^3 + 3x^2 + 4x + 5)$$